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CHM1001 2025 term 1

Assignment 6

1. Multiple Choice Questions (Only one correct answer per question)

1. Which is a state function?
   a) Heat (q)
   b) Work (w)
   c) Internal energy (E)
   d) Path length

2. In the first law $\Delta E = q + w$, w is positive when:
   a) System expands
   b) Surroundings do work on system
   c) System does work on surroundings
   d) Volume is constant

3. Work in L·atm for expansion from 2.00 L to 5.00 L at 1.00 atm:
   a) +3.00
   b) −3.00
   c) +5.00
   d) −5.00

4. At constant pressure, q_p =:
   a) ΔE
   b) ΔH
   c) $-\Delta E$
   d) $-\Delta H$

5. An endothermic process:
   a) Involves negative ΔE
   b) Releases heat
   c) Absorbs heat
   d) Involves negative ΔH

6. Calorimetry in a coffee-cup measures:
   a) ΔE at constant volume
   b) ΔH at constant pressure
   c) ΔE at constant pressure
   d) ΔH at constant volume

7. The bomb calorimeter measures:
   a) Heat at constant P
   b) Heat at constant V
   c) Latent heat only
   d) Work directly

8. In Hess's Law, ΔH is:
   a) Path dependent
   b) Path independent
   c) Proportional to path length
   d) Undefined

9. $\Delta H°f$ of elements in their standard state is:
   a) 0 kJ/mol
   b) 1 kJ/mol
   c) 100 kJ/mol
   d) Varies

10. $\Delta H°rxn$ from $\Delta H°f$ data:
   a) $\Sigma\Delta H°f(\text{products}) - \Sigma\Delta H°f(\text{reactants})$
   b) $\Sigma\Delta H°f(\text{reactants}) - \Sigma\Delta H°f(\text{products})$
   c) Average of reactants and products
   d) None

11. Conversion 1 L·atm = ? J
  a) 101.3
  b) 4.184
  c) 8.314
  d) 273

12. In $\Delta E = q + w$, for adiabatic expansion of an ideal gas:
  a) $q > 0$
  b) $q = 0$
  c) $w = 0$
  d) $\Delta E > 0$

13. Increasing reactant amounts by factor n changes ΔH_{rxn} by:
  a) $\Delta H/n$
  b) $n\Delta H$
  c) No change
  d) Cannot predict

14. Which statement about spontaneous processes is correct?
  a) Always fast
  b) Always exothermic
  c) Occur without continuous external input once started
  d) Have $\Delta S = 0$

15. For a reversible process:
  a) $\Delta S_{univ} > 0$
  b) $\Delta S_{univ} = 0$
  c) $\Delta S_{univ} < 0$
  d) $\Delta S_{sys} = 0$ always

16. An increase in entropy is observed for:
  a) Freezing water
  b) Condensation

c) Vaporization
d) Crystallization

17. The second law of thermodynamics states:
a) $\Delta E\_univ > 0$
b) $\Delta S\_univ > 0$ for spontaneous processes
c) $\Delta H\_univ > 0$
d) $\Delta G\_univ > 0$

18. Which has the greatest standard molar entropy at 298 K?
a) $H_2O(s)$
b) $H_2O(l)$
c) $H_2O(g)$
d) All the same

19. Standard molar entropy values are:
a) Zero for all substances at 0 °C
b) Always positive above 0 K
c) Negative for gases
d) Larger for elements than compounds

20. For the reaction: $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$, $\Delta S°$ is:
a) Positive
b) Negative
c) Zero
d) Cannot tell

21. $\Delta G < 0$ means:
a) Spontaneous at given conditions
b) Nonspontaneous
c) At equilibrium
d) Endothermic

22. If $\Delta H < 0$, $\Delta S > 0$, reaction is:

a) Spontaneous at all T
b) Nonspontaneous at all T
c) Spontaneous only at high T
d) Spontaneous only at low T

23. If $K = 1$, ΔG° =:

a) Positive
b) Zero
c) Negative
d) Undefined

24. If $\Delta G^\circ = -10$ kJ/mol at 298 K, K is:

a) >1
b) <1
c) =1
d) Cannot tell

25. Entropy change for surroundings at constant P:

a) $\Delta S_{surr} = -\Delta H_{sys} / T$
b) $\Delta S_{surr} = \Delta H_{sys} / T$
c) $\Delta S_{surr} = \Delta H_{sys} / P$
d) $\Delta S_{surr} = \Delta E_{sys} / T$

26. Third law says:

a) S of perfect crystal = 0 at 0 K
b) S of perfect crystal = finite at 0 K
c) $\Delta S = 0$ for all processes
d) S is independent of T

27. Highest ozone concentration is in the:

a) Troposphere
b) Stratosphere
c) Mesosphere
d) Thermosphere

28. Ozone forms when:

a) $O_2 \rightarrow 2O$ by UV light, O combines with O_2
b) O_2 absorbs IR
c) O_3 absorbs UV-B
d) N_2 dissociates

29. CFCs deplete ozone because they:

a) Absorb UV directly
b) Release Cl· radicals upon photolysis
c) Increase CO_2
d) Produce SO_2

30. Acid rain from SO_2 involves intermediate formation of:

a) H_2SO_4 directly
b) H_2O_2
c) SO_3
d) H_2S

31. Greenhouse gas with highest present-day atmospheric concentration is:

a) CO_2
b) CH_4
c) N_2O
d) CFC-12

32. Eutrophication in lakes is caused mainly by:

a) Oil spills
b) Nutrient runoff (nitrates, phosphates)
c) Thermal pollution
d) PCB contamination

33. Increased CO_2 affects ocean chemistry by:

a) Increasing pH

b) Decreasing pH (acidification)
c) Creating alkaline water
d) Removing bicarbonates

34. Which of the following is not the 12 principles associated with green chemistry?
a) chemical products should be designed to minimize toxicity even at the cost of their desired function 4
b) the end products of chemical processing should break down at the end of their useful lives into innocuous degradation products 10
c) promotion of the use of catalysts 9
d) inherently safer chemistry for accident prevention

35. The visible reddish-brown component of photochemical smog is:
a) NO_2
b) O_3
c) PAN
d) CH_4

36. Which step in water treatment kills pathogens?
a) Sedimentation
b) Filtration
c) Chlorination or ozonation
d) Aeration

2. Short-Answer/Calculation Questions

1. A steel cylinder absorbs 15.5 kJ heat and does 4.2 kJ work. Find ΔE.
$\Delta E = q + w = (15.5\ kJ) + (-4.2\ kJ) = 11.3\ kJ$
2. For reaction with ΔH°f data: $H_2O(l)$: −285.8 kJ/mol, $H_2(g)$: 0, $O_2(g)$: 0.
Find ΔH° for $2H_2 + O_2 \rightarrow 2H_2O(l)$. $\Delta H^\circ_f = H_{product} - H_{reactant} = 2(-285.8) - 2(0) - 0 = -571.6\ kJ/mol$
3. Calculate ΔS°rxn at 298 K for $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, given
S°(N_2)=191.5, S°(H_2)=130.6, S°(NH_3)=192.8 J/mol·K. $\Delta S^\circ = \sum S^\circ_{products} - \sum S^\circ_{reactants}$
$= 2(192.8) - (191.5 + 3 \cdot 130.6) = -197.7$ J/mol·K
4. Determine spontaneity: ΔH = −50 kJ, ΔS = −100 J/K at 298 K.
$\Delta G = \Delta H - T\Delta S = -50\times10^3 - 298(-100)$
$= -20200\ J$ → Spontaneous in forward direction
5. Consider the reaction:

$Ag^+ (aq) + Cl^- (aq) \rightarrow AgCl (s)$

Given the following table of thermodynamic data at 298 K:

Substance	ΔH_f° (kJ/mol)	S° (J/K· mol)
Ag^+ (aq)	105.90	73.93
Cl^- (aq)	-167.2	56.5
AgCl(s)	-127.0	96.11

$\therefore \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -65.7 \times 10^3 - 298 \cdot (-34.32)$ J/mol

≈ -55472.64 J/mol

$K = e^{-\Delta G^\circ/RT} = e^{-\left(\frac{-55472.64}{8.31 \cdot 298}\right)} = 5.35 \times 10^9$

Find the value of K for the reaction at 25 °C.

$\Delta H^\circ_{rxn} = -127.0 - (-167.2 + 105.90) = -65.7$ kJ/mol

$\Delta S^\circ = 96.11 - (56.5 + 73.93) = -34.32$ J/mol·K

6. Explain how Cl· from CFCs destroys ozone catalytically.

$2Cl(g) + 2O_3(g) \rightarrow 2ClO(g) + 2O_2(g)$ (Chlorine from CFC)

$2ClO(g) + h\nu \rightarrow 2Cl(g) + 2O(g)$ (Photodissociation of ClO)

$O + O \rightarrow O_2$

$\cancel{2Cl(g)} + 2O_3(g) + \cancel{2ClO(g)} + \cancel{2O(g)} \rightarrow \cancel{2Cl(g)} + \cancel{2ClO(g)} + 3O_2(g) + \cancel{2O(g)}$

$2O_3(g) \xrightarrow{Cl} 3O_2(g)$

→ Cl activates the conversion of O_3 into O_2, which destroys and decrease the amount of O_3